Maria Paul Thurkadayil

ASPIRING DATA SCIENTIST

Contact

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Skills

Generative AI: Prompt Engineering, Hugging Face, Transformers
Deployment, Containerisation Docker, TailScale, SkyPilot, Flask Streamlit,
DBMS: MySQL, PostgreSQL, MongoDB, MariaDB
Analytics & Visualisation: Python, Power BI, MS-Excel
Data Science Libraries: Pytorch, Keras, sklearn, pandas,
Programming: Data Versioning, HTML,CSS, ,
Statistical Analysis, Unit Testing/Debugging, Prompt Engineering, LLM Evaluation.

Certifications

Google IT Support Certification,
Introduction to Linux,
SQL for Data Science,
Machine Learning Foundations: A Case Study Approach ,
Crash Course on Python

Teaching

Teaching Assistant, NPTEL
(Jan 2024- April 2024)
noc24-cs47:

The Joy of Computing Using Python
Prepared weekly assignments & handled the forum for the course having around 60,000 registered learners.

Student Mentor, IITM BS Degree Programme
(Mar 2024- April 2024)
BSMA1004:

Statistics for Data Science 2.
Conducted doubt clearing sessions of 2 hours, 3 times a week for a set of 25 students

Summary

Quantitatively trained Data Scientist with a Master's from CMI . Strong foundation in statistics, probability, and Python. Proven ability to design AI evaluation workflows, write rigorous tests to verify model outputs, and document complex assumptions. Seeking to leverage analytical rigor and AI-systems knowledge as a Quantitative Research and AI Systems Associate."

Experience

Project Intern, IIT Jammu | June 2026- Present

Quality-Clustering

- Developed statistical validation framework for quality-bucketing of large text corpora. Implemented topic erasure via linear residualization

Multilingual LLM Evaluation Pipeline

- Built Kubernetes-based evaluation infrastructure (V100 GPU cluster, SkyPilot orchestration) for large-scale multilingual LLM assessment.
- Generated reproducible artifact pipeline:
technical reports + interactive HTML visualizations with embedded plots and sample-level data.

Gaperon mmbert distill repository

- Engineered end-to-end ML pipeline (HuggingFace Hub integration, SkyPilot orchestration) for multilingual model distillation: mmBERT-base → 6-language classifier with independent heads (clarity, coherence, depth, grammar, usefulness, overall). optimized for cost and latency across 6 languages (English, Hindi, Tamil, Bengali, Telugu, Malayalam).

Summer Research Intern, S3 Labs, IIT Bhiliai | May 2025 - July 2025

1. **Technical Image Understanding**
- Evaluating how Packet Diagrams, Sequence Diagrams, Class Diagrams, State Diagrams, etc are inferenced using LLM's..
2. **Legal Infographic Generation**
Dataset : IL-TUR(Exploration Lab, IIT Kanpur)
- Conducted Research Paper Analysis on the Applicability of LLM's in the Legal Domain
- Generated trial prototypes for generating Infographics like Timelines, Key Arguments
- **Supervisor: Gagan Raj Gupta, Department of Computer Science and Engineering**

Projects

1. **Forecasting the Total Marine Fish Production In Kerala - Development & Comparative Analysis of Machine Learning Models**

Machine Learning, Fisheries | Jan 2022- May 2022

- Marine Fish production Data in Kerala (in tonnes) was collected from **CMFRI, Kochi** and data on 9 significant environmental variables from **ICOADS** from 1960 - 2014 .
- Highest predictive power - Random Forest- R2 score 0.87, MAE 0.05, MSE 0.005, RMSE 0.07

Volunteering

Practical Agentic AI Workshop Series by Prof Raghav Kulkarni, CMI |
August 2026- Present

Conducting Sessions every Sunday from 4PM to 7 PM on Agentic AI.

Coder's High Volunteer :
Volunteered for Workshop Series by Sudarshan Iyenger, IIT Ropar

COVID-19 Response Volunteer:
Caregiver Saathi, May 2021

Hobbies

Room Recreation, Reading.

Others

.GATE DA 2025: Qualified
(March 2025)

Winner Hack Kochi #2:
(November 2019)
IEEE ORE, Kochi Hub

KEAM- Engineering: Rank 333
(August 2018)
Kerala State Entrance Examination

Highest Scorer: Science Stream:
Subject Topper for Maths, Chemistry and Computer Science
(May 2018)

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2. VR Industrial Training Simulator | Unity, C# | Jan 2021
Co-developed a Virtual Reality application simulating boiler feedwater plant control room operations, creating a safe, immersive environment for industrial operator training Collaborated on engineering interactive 3D scenarios and automation workflows in Unity/C#, effectively eliminating the physical risks and costs associated with traditional models.

3. . Prediction of Crime Categories | Ensembled Learning, Law & Order | August 2024

- The dataset consisted of 20,000 rows & 20 columns about location, victim demographics, date, time, etc. 6 crime categories were possible for the target variable.
- Data Exploration, Cleaning and Preprocessing created 23 relevant feature columns.
- **LGBMClassifier()** and **XGBoostClassifier()** were used in building a **VotingClassifier()** for which had an accuracy of 0.88080 in contrast to the baseline accuracy of 0.58

4. Perspective-Aware Sentiment Analysis | Applied Data Analytics | November 2025

- Built a real-time data pipeline using **Google News RSS** to fetch headlines and established an objective sentiment baseline using the **cardiffnlp/twitter-roberta-base-sentiment** model.
- Evolved the initial single-prompt **GPT-4o-mini persona engine** into a **multi-agent debate framework using LangGraph**, orchestrating s**pecialized agents (Advocate, Context Expert, Skeptic, Judge)** for adversarial reasoning.
- Transformed static polarity labels into explainable, perspective-aware sentiment scores, significantly reducing hallucinations and capturing deep subjective nuances that traditional models miss.

5.. Opacus - Differential Privacy in PyTorch
Open Source Contribution, Data Privacy and Data Quality Project | May 2026

- Attempted a Contribution to Opacus, Meta's open-source library enabling differentially private deep learning with minimal code changes. Worked on an issue that had a privacy bug (silent epsilon miscalculations with WeightedRandomSampler)

6. Structured Multi-Agent Pricing Engine | Agentic AI | November 2026

- Built a real-time conversational pipeline using the **OpenAI Agents SDK** and **Google Gemini API** to automate e-commerce price discovery, orchestrating specialized agents (Triage, Researcher, Calculator) for seamless task routing.
- Evolved the initial single-agent prompt into a multi-agent orchestration framework using **Pydantic** for strict schema validation and **SQLite** for persistent session memory, ensuring type-safe data handoffs and context-aware interactions.
- Transformed unstructured user queries into structured, actionable pricing records, significantly reducing model hallucinations and preventing infinite tool loops through rigorous prompt constraints and state-tracking.

Education.

Chennai Mathematical Institute, **Master of Science**
Data Science, Aug 2024 - May 2026